Science & Technology Facilities Council
Daresbury Laboratory

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A Polarity Reversal Option for the Quadrupole and Dipole magnets

to the European Collaboration for the

Gamma-Beam System to the

Extreme Light Infrastructure – Nuclear Physics (ELI-NP) Project

Version 1.0

Document Approval

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Introduction

STFC has been requested by EuroGammaS to provide an option for a solution to degauss all 36 quadupole magnets and 2 dipole magnets (LEL-MAG-DIP01 and LEL-MAG-DIP02).

Degaussing magnets has the benefit of a guaranteeing a reproducible magnet field, within tolerances specified.

To implement the system for the 38 magnets, 29 modules will be required and additional rack space.

- 50A units for Quadrupoles type D & E, quantity 10
- 100A units for Quadrupoles type B, C & F, quantity 17
- 200A units for Dipoles type A & B, quantity 2

System Description Overview

The power supplies specified and currently being procured for the ELI-NP quadrupole and dipole magnets are unipolar units and can only operate in two quadrants. To enable the magnets to be degaussed the power supplies must be bipolar and operate in four quadrants. Unfortunately the cost of procuring true bipolar power supplies for these types of magnet is expensive due to the significantly higher power ratings than corrector magnets.

A low cost alternative is to utilise change-over DC contactors with silver alloy tips, which are weld resistant, hard wearing and have excellent conductivity. These will allow a degauss cycle to be completed by initially cycling the magnet with positive current and once current returns to zero operate the change-over switch and cycling the magnet with negative current. This process can be repeated as required by the user with maximum currents dictated by either the contactor or power supply ratings.

The switching is performed at zero current to maximise the life of the contacts. Contactors are magnetically latched so that they remain in the selected position even in the event of power failure to the contactor crate.

Two contactors are provided for each channel. Each has auxiliary switches fitted to indicate the position of the contactor.

Control of the contactors is via external volt free contacts (relays); applying a signal to ether the Normal or Reverse control pin will cause the contactors to change polarity. The control pulse to the contactor coils is generated within the contactor crate. LEDS are provided on the front panel to indicate the status of the contactors. This status is also indicated remotely to the control system via the rear panel connectors.

Sequencing of the contactor operation is carried out by an internally mounted programmable relay. This can be configured for either individual channel control where there are separate signals from the control system for each channel or group control where several reversing crates can be operated from a common control signals. A 2-channel reversing crate is provided for magnet currents up to 60 A. This occupies 2U of rack space.

For higher currents a single channel is provide per crate requiring up to 3U of rack space.

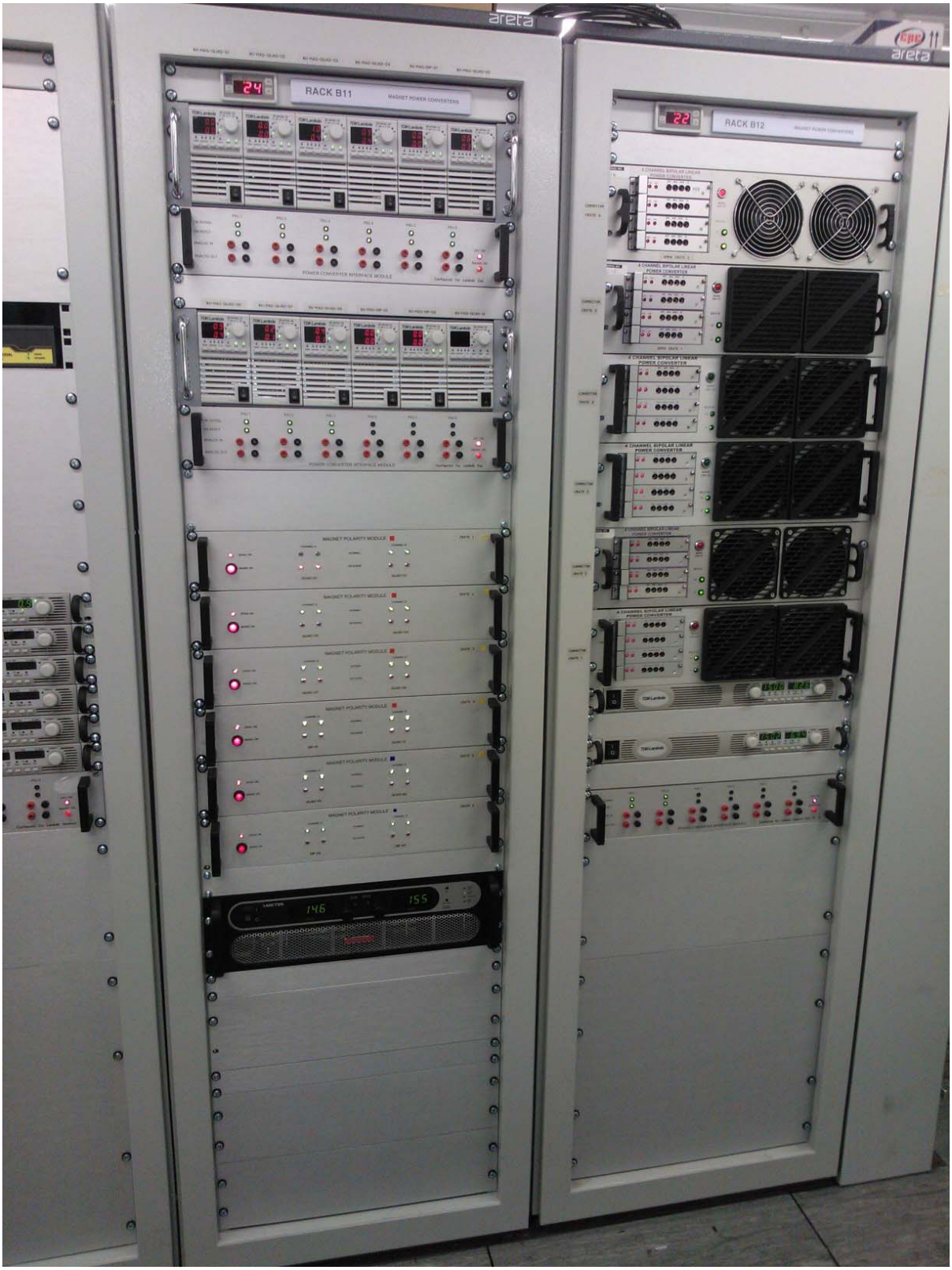

Figure 1: Magnet Power supply racks (Low Energy) Front

Figure 2: Example of 60 A polarity reversal equipment modules implemented on ALICE @ Daresbury

Cost

The total cost to implement this option for hardware and staff is €96,623.